

Welber S. de Souza

Aeronautics Institute of Technology (ITA)

Praça Marechal Eduardo Gomes, 50, São José dos Campos, SP, 12228-900, Brazil

welber@ita.br • +55 (11) 98653-1122 • [linkedin.com/in/welber-souza](https://www.linkedin.com/in/welber-souza)

Education

Aeronautics Institute of Technology (ITA), São José dos Campos, SP, Brazil

Expected July 2027

M.S. in Aeronautical and Mechanical Engineering - Aerospace Structures

- Thesis: *Semi-analytical finite element modeling of riveting-induced plastic deformation in aeronautical structures*. Advisor: Dr. Alfredo Rocha de Faria
- Grade: 9.06/10 (estimated GPA 3.80/4.0). Selected coursework: Fundamentals of Elasticity and Plasticity (9.8); Finite Elements I (8.6); Computational Modeling in Engineering (8.3); Plate and Shell Theory (in progress)

Getúlio Vargas Foundation (FGV), Rio de Janeiro, RJ, Brazil

2022

Specialization in Project Management (grade: 9.19/10)

Federal Institute of São Paulo (IFSP), São Paulo, SP

2018

Bachelor of Science in Civil Engineering (grade: 7.18/10)

- Thesis: *EDUCASYS: a finite element mesh generator for building floor systems*. Advisor: Dr. Edson Bispo Ferreira

Federal Institute of São Paulo (IFSP), São Paulo, SP

2012

Technical Degree in Building Construction

Additional training: Computational Fluid Dynamics Applied to Tall Buildings, short course, IFSP (Feb-Jun 2026)

Research Experience

Aeronautics Institute of Technology (ITA), São José dos Campos, SP

2025 - Present

Graduate Researcher; Advisor: Alfredo Rocha de Faria

My research focuses on how manufacturing processes induce localized nonlinear effects, such as plasticity and residual stresses, that govern the global performance of structures, and on efficient computational methods that capture these effects without resorting to full nonlinear analyses. My work combines closed-form analytical solutions, custom finite element formulations and validation against high-fidelity simulations and experimental data.

- Derived an analytical elasto-plastic hole-expansion solution (von Mises yielding, Hollomon hardening) for the plastic zone and displacement field around the rivet hole.
- Formulated special plate elements with an embedded rivet that transfer this local solution as prescribed displacements to a linear-elastic global mesh, avoiding a full nonlinear analysis of the panel.
- Ongoing: implementation in the research group's in-house finite element code, validation against nonlinear finite element analyses and experimental data, and application to the effect of riveting sequence on residual stresses and panel distortion.

Federal Institute of São Paulo (IFSP), São Paulo, SP

Oct 2016 - Dec 2018

Undergraduate Researcher; Advisor: Edson Bispo Ferreira

- Developed a CAD-based modeling interface with discretization modules in Fortran, Python and AutoLISP for the finite element analysis of multi-story building floor systems (basis of the B.S. thesis EDUCASYS).

Federal Institute of São Paulo (IFSP), São Paulo, SP

Jun 2013 - Jun 2014

Undergraduate Researcher, CNPq "Young Talents for Science" Fellow; Advisor: Fabio Maluf

- Developed Java software for internal forces in simply supported beams under arbitrary loading, and an image-based center-of-mass locator using pixel-wise scanning and matrix processing.

Publications and Presentations

Conference Papers

All papers were presented orally (* presenting author).

- **Souza, W. S.***, Marques, D. S.*, Silva, C. C. (2026) Computational modeling of diaphragm walls via soil-structure interaction in Python and Matlab, *22nd COBRAMSEG - Brazilian Congress of Soil Mechanics and Geotechnical Engineering*.
- Guimarães, A. L. A., Cardoso Jr., E. A., Marques, D. S., **Souza, W. S.*** (2026) Application of network topological analysis to the evaluation of a proposed structuring axis for the East Zone of the São Paulo metro network, *AEAMESP - 32nd Metro-Railway Technology Week*, São Paulo.
- Marques, D. S.*, **Souza, W. S.***, Silva, E. J. (2025) INSPECTUS: Route optimization tool for rural aerodrome inspection using the Traveling Salesman Problem, *22nd SITRAER - Air Transportation Symposium*, Brasília. doi: [10.29327/9786527220244.1239778](https://doi.org/10.29327/9786527220244.1239778)
- **Souza, W. S.***, Jazra, F. M. (2013) Development of a didactic application for internal forces in simply supported planar beams, *5th CINTEC - Scientific Initiation Congress of the Federal Institute of São Paulo*, São João da Boa Vista.

Technical Magazine Articles

- Silva, I. M., Kenchian, A., **Souza, W. S.**, Siqueira, R. G. (2015) A proposal for the gateway of the city of Bonito-MS, *Concreto & Construções*, 1, pp. 142-150.

Books

- **Souza, W. S.**, Marques, D. S. High school physics textbook (in Portuguese, in preparation). Developed from over a decade of volunteer physics teaching to approximately 1,000 pre-university students.

Invited Talks

- **Souza, W. S.**, Bucker, M. B. (2025) Evolution of São Paulo Metro construction works and challenges in structural design, *SEDICITEC - Science and Technology Week*, IFSP, São Paulo.

Manuscripts in Preparation

- **Souza, W. S.**, Marques, D. S., Silva, E. J., Rocha de Faria, A. INSPECTUS: An intelligent decision support system for aerodrome inspection route optimization.
- **Souza, W. S.**, Bussamra, F. L. S., Ferreira, E. B., Rocha de Faria, A. Open-source CAD-CFD-FEM pipeline for the aerodynamic loading of airfoils: structured mesh generation and validated one-way coupling.

Teaching and Mentoring

Federal Institute of São Paulo (IFSP), São Paulo, SP

2026 - Present

Co-advisor, Undergraduate Final Thesis (student: Aislan Paes; advisor: Prof. Cátia C. Silva)

- Guiding the development of a finite element processor for the plane-strain analysis of circular and oval underground openings (tunnel cross-sections), with results benchmarked against PLAXIS.

SEDICITEC - Science and Technology Week, Federal Institute of São Paulo (IFSP), São Paulo, SP

2025

Short Course Instructor

- Designed and taught *Analysis of Diaphragm Walls via the Finite Difference Method using Python and Matlab*, a 6-hour hands-on course for 10 participants.

Bonsucesso Community Prep Course (CCB), São Paulo, SP

2015 - Present

Volunteer Physics Teacher

- Teaching physics to pre-university students from low-income communities, preparing them for university entrance exams; approximately 1,000 students taught over the period.
- Developing, from this experience, a high school physics textbook (see Books).

Federal Institute of São Paulo (IFSP), São Paulo, SP

Spring 2015

Undergraduate Teaching Assistant, Strength of Materials II (UG); Supervisor: Prof. Cátia C. Silva

- Teaching assistant to the undergraduate course Strength of Materials II. Topics included stress and strain transformation, combined loading, beam deflections, column buckling and energy methods.
- Held review sessions on course concepts and homework problems, and provided one-on-one support to students preparing for exams.

Professional Experience

INGENIA Engenharia e Gerenciamento, São Paulo, SP

Mar 2013 - Present

Managing Partner and Technical Director (partner since 2019)

2020 - Present

- Manager of the independent technical verification contract for the São Paulo Metro Line 2-Green Phase 2 expansion (Lots 2, 6, 7 and 8, Ponte Grande Station, Tamanduateí Yard and Anália Franco substation): technical review of structural calculation reports for reinforced concrete, steel and deep foundation systems, including finite element models and Brazilian code verification.
- Developed automated model verification workflows: IFC/BIM auditing with buildingSMART IDS 1.0 and Python (ifcopenshell), and a web application for document review workflows.
- Contract IC 1001654701, Line 2-Green (Lots 1, 5 and 6): supervision of underground infrastructure works with TBM and NATM excavation.
- Wind loading analysis to ABNT NBR 6123 for rooftop photovoltaic installations on hospitals and schools, including horizontal lifeline systems.

Project Coordinator

Apr 2018 - 2020

- Construction supervision of the Line 17-Gold monorail works, São Paulo Metro.
- Executive structural design for the Jacareí wastewater treatment plant expansion, a stormwater pumping station and underground drainage galleries (federal PAC sanitation program).

Structural Designer

May 2016 - Mar 2018

- Executive structural design and finite element modeling of elevated monorail stations for the Line 15-Silver, São Paulo Metro: access buildings of Iguatemi Station and external structures of São Mateus, Fazenda da Juta, Jardim Planalto, Sapopemba and Jardim Colonial stations.
- Design of steel pedestrian footbridges spanning 40 m and of 15 m cantilevered column corbels supporting the station structures.

Building Construction Technician

Mar 2013 - Apr 2016

- Line 5-Lilac expansion, São Paulo Metro: participation in the review and approval of design documents, and independent technical audit of physical and financial progress with deviation analysis.
- Structural, steel and foundation design for the ETEC Elias Nechar technical school (Catanduva, SP).

Independent Structural Design Consultant, São Paulo, SP

2019 - 2020

International project

- Extension block of the Santa Casa da Misericórdia de Vila Franca de Xira, Portugal (with Paes Mamede): concrete foundations, superstructure and steel façade, delivered in BIM.

Awards and Honors

- “Alumni Who Inspire” Award (*Egressos que Inspiram*), first edition, Civil Engineering Department, IFSP, at SEDICITEC 2025
- Best and Largest Works Award, Instituto de Engenharia - Line 2-Green expansion from Vila Prudente to Penha, São Paulo Metro 2024
- Academic Honor Diploma, IFSP 2015

- First place, 8th Boldness (*Ousadia*) Design Competition, a national competition held by IBRA-CON (Brazilian Concrete Institute) at the 58th Brazilian Concrete Congress 2015
- Undergraduate Research Fellowship, “Young Talents for Science” Program, CNPq 2013 - 2014

Media Coverage

- *Campus São Paulo realiza projeto “café com o egresso”* [São Paulo campus holds “Coffee with Alumni” project; featured alumnus honoree]. IFSP institutional news, September 25, 2025. ifsp.edu.br/unidades/17-ultimas-noticias/5391

Technical Skills

- **Methods:** finite element and finite difference methods; in-house finite element code development; nonlinear and elasto-plastic analysis; structural dynamics; computational fluid dynamics.
- **Programming:** Python, Fortran, C++, Java, MATLAB, AutoLISP, $\text{\LaTeX}$.
- **Software:** CalculiX, OpenFOAM, SAP2000, PLAXIS, STRAP, TQS, Solibri, ifcopenshell, Revit, Navisworks, Civil 3D, AutoCAD.
- **Software developed:** DIMENSIONA (2016 - present), Java program for the structural design of plates and bars, integrated with finite element software and structural detailing; Beam Solve (2014); Center of Mass Locator (2013).

Professional Memberships

- Instituto de Engenharia, São Paulo, Brazil - Member 2024 - Present
- Regional Council of Engineering and Agronomy of São Paulo (CREA-SP), registration 5070218785 2018 - Present

Service and Outreach

Federal Institute of São Paulo (IFSP), São Paulo, SP

2016 - 2018

Volunteer, Technical Assistance Program for Low-Income Housing

- Member of a volunteer design team in partnership with a non-governmental organization, coordinated by Dr. Alexandre Kenchian.